Tragedy of the Commons

Facilitation Materials – Step-by-Step Guide

This **Step-by-Step Guide** is intended to give a detailed, procedural view on how to create and run a game of ToC for teachers who have already read the Introductory Guide.

How to Create a Game

Necessary materials:

- Reliable wireless Internet that all players' and facilitators' devices can access
- A chalk/dry-erase board for recording discussion notes
- A means of sharing the Game Code, URL, and other set-up steps for players (e.g. a handout, digital slide, or space on a chalk/dry-erase board)
- A functional printer and paper with which to print QR codes for the game stations

For teachers

1. Use any device (e.g. phone, tablet, laptop, or desktop computer) that can connect to an Internet browser.
    - A tablet or desktop is recommended.
2. Go to **psims.games** and select **Host a Game.**
3. Log in with a Google account.
    - Tip: If another person will be facilitating the game (e.g. a co-teacher), consider creating a shared Google account that others can access.
4. Either **Create New Game** or **Choose Existing Game.**
    - If creating a new game, select ToC from the **Choose a Game Type** dropdown and create a unique **Game Code**.
    - If choosing an existing game, select the **Game Code** corresponding to the desired game, and press **Choose Game.**
5. Write the **Game Code** on the board for students to enter and join the game.

For students

1. Use a charged, WiFi-connected mobile device with a camera and:
    - Android devices: Chrome browser
    - Apple devices [iOS 11 or later]: Safari browser
    - Fire tablets: Silk browser
2. Go to **psims.games** and select **Join a Game.**
3. Enter the **Game Code** (created by the host).
4. Enter a **Display Name** and select **Join.**
5. At the **Enter Game** prompt, select **Continue.**

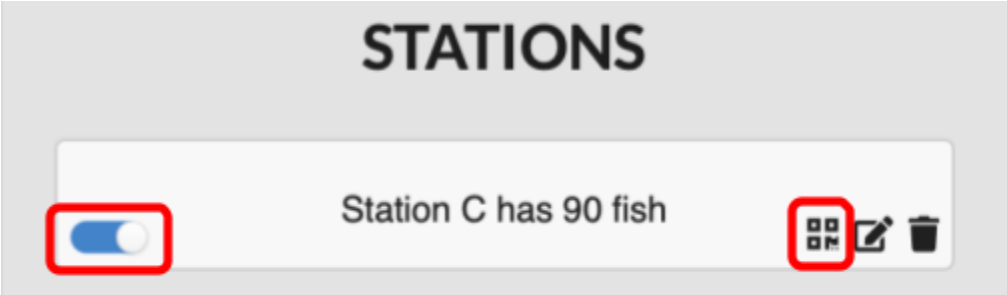


How to Set Up the Game

After creating the game, the facilitator will see the setup dashboard with the following circled options:



1. Stations	Add or remove stations where players can fish, name them, and choose the starting number of fish in each of them. **Create New Fishing Station** **Station Name** Station C **Starting Fish Amount** 90 0 500 Create Cancel

<table>
<tr>
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After a station is created, the dashboard displays it with these circled options:


Toggle the switch in the bottom left corner to disable or enable the station. This can only be done during initialization, before the game has been launched. Disabled stations won't show up in any graphs.

Select the QR code icon in the bottom right corner beside each station to view and print out its corresponding QR code. The QR codes for disabled stations are no longer scannable.

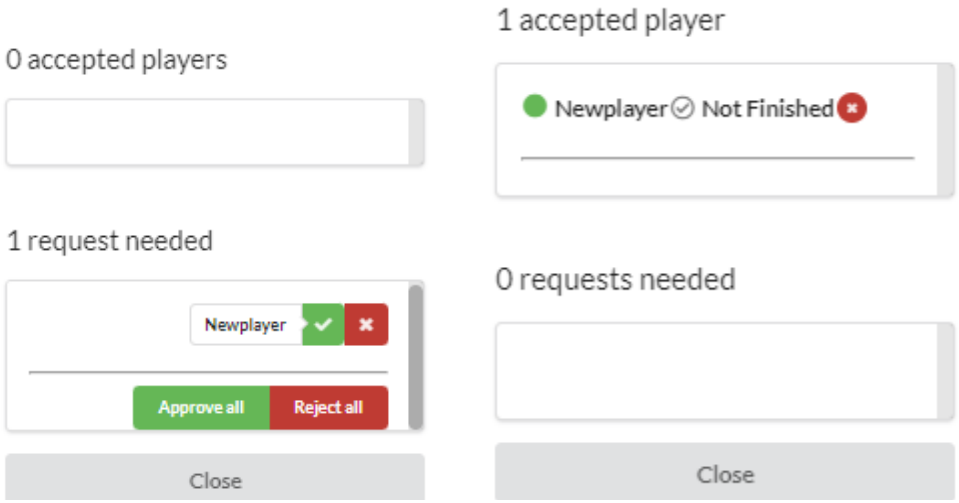
Display all printed station QR codes around the room, spreading them out. It is recommended to use the following combinations of number of station and starting number of fish:

<table>
<tr><th># of players</th><th>Recommended starting setup</th></tr>
<tr><td><= 5</td><td>3 stations, ~50 fish each</td></tr>
<tr><td>6-10</td><td>3 stations, ~100 fish each</td></tr>
<tr><td>11-15</td><td>5 stations, ~150 fish each</td></tr>
<tr><td>16-20</td><td>5 stations, ~200 fish each</td></tr>
<tr><td>21-25</td><td>7 stations, ~200 fish each</td></tr>
<tr><td>26-30</td><td>7 stations, ~250 fish each</td></tr>
<tr><td>31-40</td><td>9 stations, ~250 fish each</td></tr>
<tr><td>41-50</td><td>10 stations, ~300 fish each</td></tr>
</table>

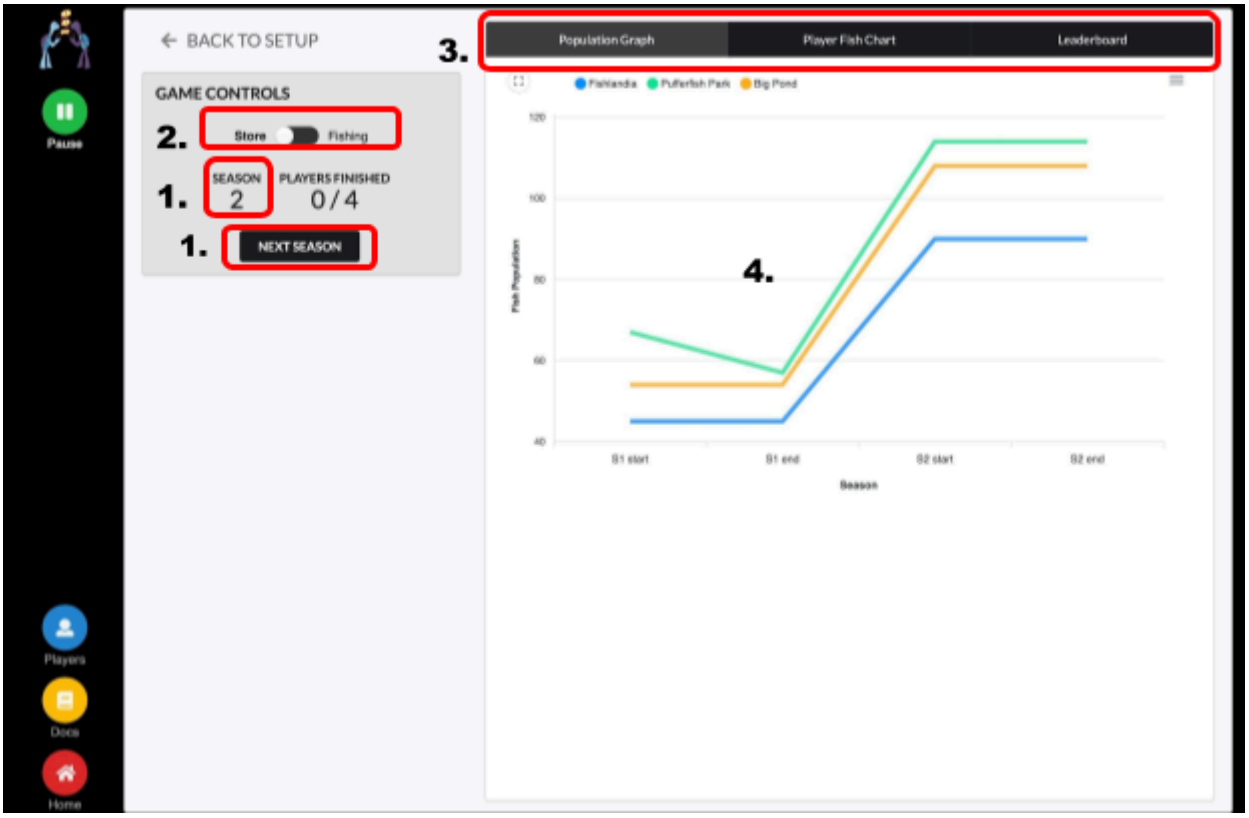
These recommended starting fish amounts are mean values. Stations should have some variability.

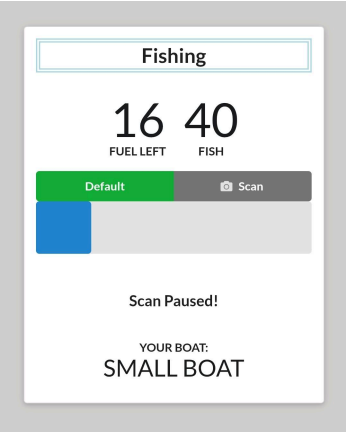
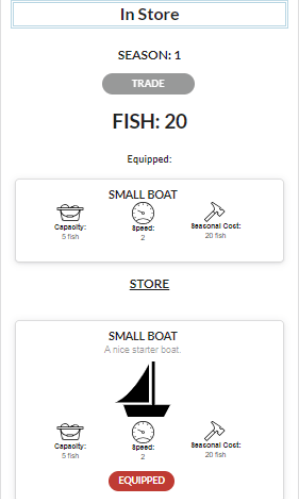
Be mindful of classroom space as well when deciding how many stations to create and how to organize them around the classroom.
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<tr>
<td>2. Local/ remote gamemode</td>
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Toggle determines whether the game is played local (face-to-face) or remote (online). This guide is focused on the local gamemode. For details about the difference between the two versions, see the Remote Documentation.

After the game has been launched, this toggle can no longer be changed.
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<tr>
<td>3. Trading</td>
<td>Optional toggle enable players to donate fish to each other in between seasons. This toggle can be changed after the game has been launched.</td>
</tr>
</table>

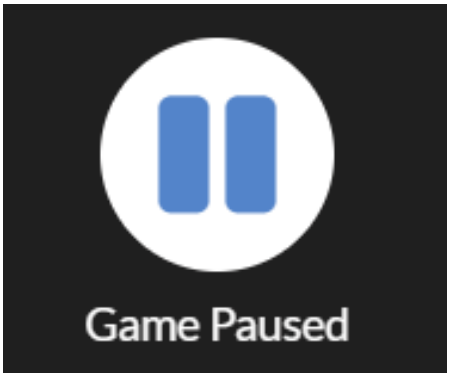
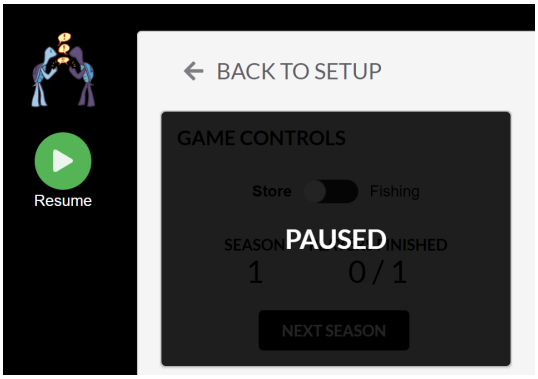
4. Central Bank	Optional toggle enables the Central Bank features. This guide assumes it is toggled off. For more details about when it is on, see the Central Bank Guide.
5. Players List	Manage players currently in the game by removing them or seeing whether they have finished fishing, and manage join requests for players not yet in the game. 

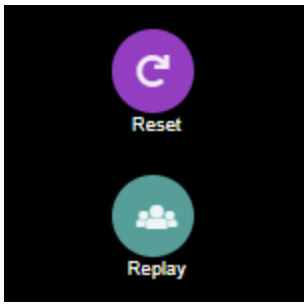
Once ready to launch the game, press **Launch Game** and then **Continue Game** to see the game dashboard with these circled options:



1. Season	The “round” of fishing that the students are in. During the season, students are focused on fishing using boats and technology. Between seasons, they can spend fish in order to upgrade or change their boats or technology. The game dashboard displays which season the game is currently in and allows the facilitator to move onto the next season.
2. Player Status	Toggle determines whether students are in the store or in fishing mode. When players are in fishing mode, they can scan station QR codes to fish in them. Each time a player fishes, they will have a five second scanning timeout and use up limited fuel in the season. When players are in the store, they can see and buy items to upgrade their boats for the cost of a certain amount of fish. If trading is enabled, they can trade here. A player screen in fishing mode vs in the store:  
3. Statistics	**Population Graph:** Displays and tracks the station populations over seasons **Player Fish Chart:** Displays and tracks players’ numbers of fish **Leaderboard:** Displays the students with the top numbers of fish, based on their current amount of fish left or their total amount of fish caught.

How to Run the Game

Start of the game	Once all students have joined the game and are ready to play: ● Ask them to get up from their seats. ● Switch the Player Status toggle from “Store” to “Fishing” to start the first season and allow students to begin fishing.
During the game	During a season, keep an eye on the game dashboard to: ● Monitor the status of the game, including the statistics and the Players Finished counter in the Game Control panel. ● Accept rejoin requests from players who log out midgame. ● Pause the game if needed by pressing the Pause button at the top of the sidebar. This pauses the game for players by preventing them from taking any actions. Student View  Teacher View  ○ The game can also be paused from the game dashboard by pressing the **Back To Setup** button in the top left corner. This allows teachers to access the initialization dashboard again and add new stations or reset/replay the game. To end the season: ● Switch the Player Status to “Store.” ● Give players sufficient time to buy items in the store and/or trade: ○ **Boats**: Upgrading and buying a bigger boat (for the cost of a certain amount of fish) will allow players to obtain more fish each time they fish and/or decrease the amount of fuel it takes to fish. ○ **Research**: Buying the research upgrade allows players to use one unit of fuel to investigate information about a station before choosing to fish in it, saving them future fuel and granting them more efficiency when fishing. ○ **Trading**: If trading is enabled, players can donate fish to other players during this time.

	After giving players sufficient time, press the **Next Season** button to move onto the next season. Then: ● Players' number of fish will be depleted by the seasonal cost of 20 fish. ○ This deduction can cause players to go into debt with their number of fish. The game is over for them if they have less than -20 fish. ○ A player's seasonal cost can change by equipping certain items from the boat store. ● The population of fish at each station will double. ● Switch the Player Status toggle from "Store" to "Fishing" to allow students to begin fishing again.
End of the game	When all (or most) of the fish are gone, end the game by: ● Not advancing to the next season. ● Asking players to take their seats. After this initial game is over, the class should have a student-driven discussion focused on the concepts and questions that players want to explore in the next game. For a more detailed explanation of how to facilitate discussion in ToC both during and after the game, see the Discussion Guide. Once the discussion is over, the game should be played again one or more times. There are two buttons in the lefthand sidebar of the initialization dashboard that the facilitator can use to quickly initiate a new game:  **Reset:** Removes all players and stations and reset all game parameters to their default values. Commonly used to start a new game with new players. **Replay:** Keeps players and stations but revert player fish amounts and station fish populations to their initial amounts. Commonly used to quickly restart while keeping everything constant.